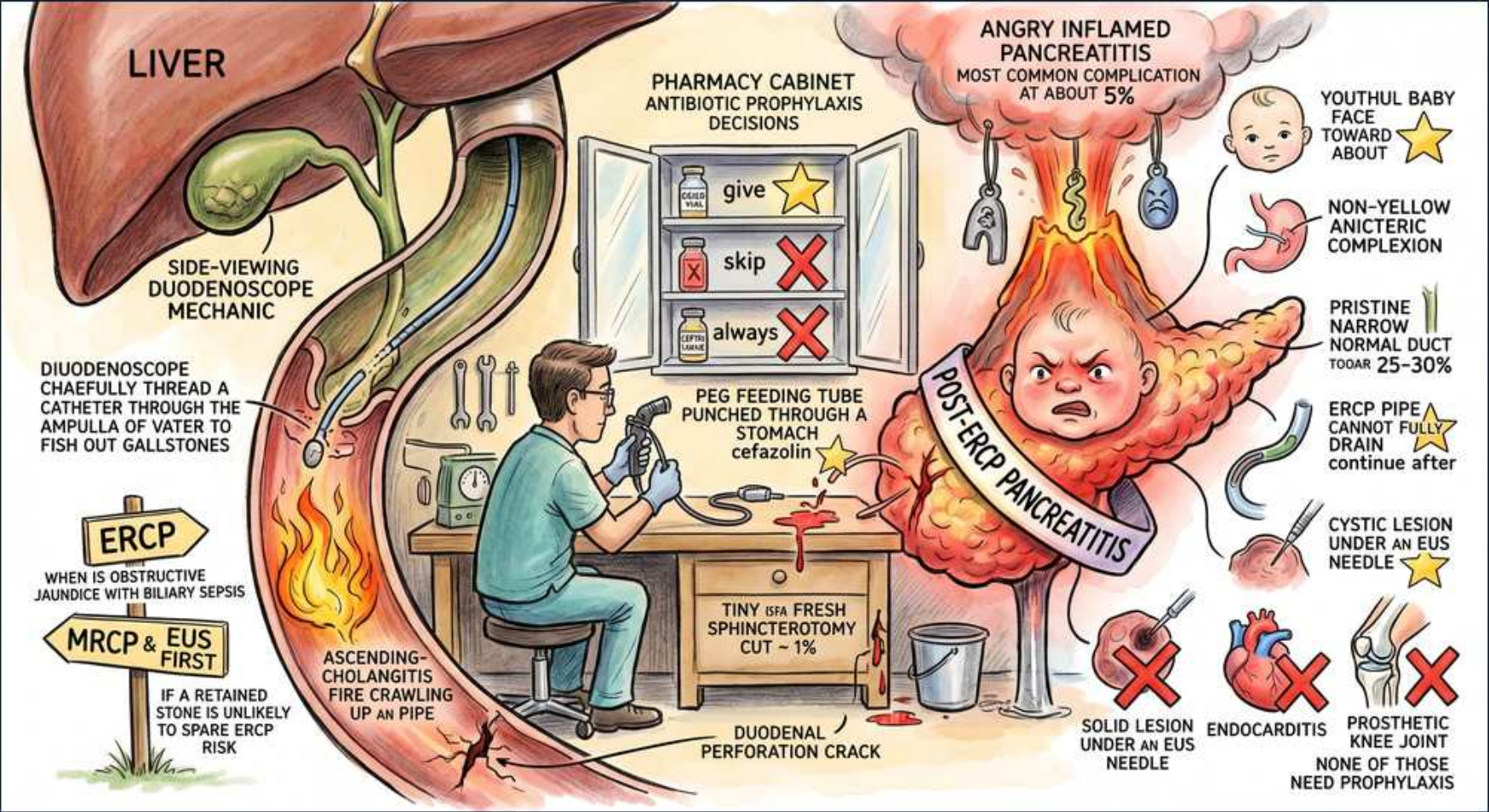


Endoscopy — ERCP Complications & Prophylaxis





1. Side-viewing duodenoscope mechanic

WHAT YOU SEE

Catheter/scope threading toward the liver, 'SIDE-VIEWING DUODENOSCOPE MECHANIC', left.

WHAT IT ENCODES

ERCP uses a side-viewing duodenoscope passed to the second part of the duodenum, cannulating the ampulla of Vater to access the biliary/pancreatic ducts and extract stones. It is therapeutic, not just diagnostic. Indicated for choledocholithiasis with cholangitis or biliary obstruction.

2. ERCP indication tag

WHAT YOU SEE

Yellow 'ERCP — when is obstructive jaundice with biliary sepsis', lower left.

WHAT IT ENCODES

Strongest indication = obstructive jaundice with ascending cholangitis/biliary sepsis (Charcot triad: fever, RUQ pain, jaundice; Reynolds pentad adds hypotension + confusion) → urgent ERCP for drainage. Also for retained CBD stones and malignant obstruction stenting.

3. MRCP & EUS FIRST (gold star)

WHAT YOU SEE

Yellow 'MRCP & EUS FIRST — if a retained stone is unlikely to spare ERCP risk', lower left.

WHAT IT ENCODES

GOLD STAR: when a CBD stone is only intermediate/low probability, do non-invasive MRCP or EUS FIRST to confirm before committing to ERCP. ERCP carries real procedural risk (pancreatitis, perforation) and shouldn't be diagnostic-only. Reserve ERCP for when intervention is needed.

BOARD TRAP

'Go straight to ERCP to look for a possible stone' → wrong; if a stone is unlikely, MRCP or EUS first spares unnecessary ERCP-related pancreatitis.

4. Post-ERCP pancreatitis volcano (~5%)

WHAT YOU SEE

Angry red pancreas erupting, 'POST-ERCP PANCREATITIS' / 'ANGRY INFLAMED PANCREATITIS most common complication at about 5%', center-right.

WHAT IT ENCODES

Post-ERCP pancreatitis is the MOST COMMON complication, ~5% overall. Caused by ampullary trauma/contrast injection. Presents with abdominal pain and elevated lipase hours after the procedure. Prevent with aggressive hydration, rectal NSAIDs, and a pancreatic-duct stent in high-risk cases.

5. Young/anicteric/narrow duct = HIGH risk

WHAT YOU SEE

Baby face 'youthful', 'non-yellow anicteric complexion', 'pristine narrow normal duct 25-30%' captions, upper right.

WHAT IT ENCODES

Highest pancreatitis risk: young, female, anicteric (normal bilirubin), normal/narrow non-dilated duct, and suspected/confirmed sphincter-of-Oddi dysfunction — risk climbs to 25–30%. Difficult cannulation and pancreatic-duct injection add risk. These patients most need prophylaxis.

BOARD TRAP

'A young woman with a normal-caliber duct is low pancreatitis risk' → wrong; that exact profile (young, female, anicteric, narrow duct, SOD) is the HIGHEST risk, 25–30%.

6. Rectal indomethacin + PD stent (prophylaxis)

WHAT YOU SEE

'ERCP pipe cannot fully drain — continue after' near pancreatitis, right.

WHAT IT ENCODES

Post-ERCP pancreatitis prophylaxis = rectal indomethacin (or diclofenac) immediately + prophylactic pancreatic-duct stent in high-risk patients, plus periprocedural IV fluids (lactated Ringer's). This is the highest-yield prevention combo on exams.

7. Ascending cholangitis fire

WHAT YOU SEE

Flames crawling up the pipe, 'ASCENDING CHOLANGITIS fire crawling up a pipe', bottom-left center.

WHAT IT ENCODES

Ascending cholangitis can follow ERCP, especially if biliary drainage is incomplete — bacteria seed the stagnant obstructed system. This is why antibiotic prophylaxis is given when drainage is expected to be incomplete.

8. Duodenal perforation crack

WHAT YOU SEE

'DUODENAL PERFORATION CRACK' caption, bottom center.

WHAT IT ENCODES

Duodenal/retroperitoneal perforation is a rare but serious ERCP complication, often related to sphincterotomy or scope trauma. Sphincterotomy itself carries ~1% bleeding/perforation risk.

9. Sphincterotomy ~1%

WHAT YOU SEE

'TINY FRESH SPHINCTEROTOMY CUT ~1%', center bottom.

WHAT IT ENCODES

Sphincterotomy carries about 1% bleeding risk. Bleeding can be immediate or delayed; managed endoscopically. Avoid in coagulopathy/antithrombotics where feasible.

10. GIVE prophylaxis — cefazolin (gold star)

WHAT YOU SEE

Pharmacy cabinet with 'give' + gold star at top shelf, center.

WHAT IT ENCODES

GOLD STAR: GIVE antibiotic prophylaxis (cefazolin/ceftriaxone) when biliary drainage will be incomplete (e.g., hilar stricture, PSC, after liver transplant) and for PEG tube placement. Also for EUS-FNA of cystic lesions. Covers seeding of obstructed/contaminated systems.



## 11. PEG tube — cefazolin

### WHAT YOU SEE

'PEG feeding tube punched through a stomach — cefazolin', center.

### WHAT IT ENCODES

PEG (percutaneous endoscopic gastrostomy) placement → give a single prophylactic dose of cefazolin to prevent peristomal wound infection. One of the few endoscopic procedures with a clear antibiotic indication.

## 12. SKIP for solid EUS-FNA (red X)

### WHAT YOU SEE

'SOLID lesion under an EUS needle' with red X, bottom center.

### WHAT IT ENCODES

EUS-FNA of a SOLID lesion does NOT need antibiotic prophylaxis. Cystic lesions are the exception (give). 'Skip' and 'always' are both wrong defaults — base it on the indication.

### BOARD TRAP

'Give antibiotics before every EUS-FNA' → wrong; solid-lesion FNA needs none — only cystic-lesion FNA gets prophylaxis.

## 13. Cystic EUS-FNA — GIVE

### WHAT YOU SEE

'CYSTIC lesion under an EUS needle', right.

### WHAT IT ENCODES

EUS-FNA of a CYSTIC lesion (e.g., pancreatic cyst) → GIVE prophylaxis (fluoroquinolone/cephalosporin) to prevent cyst infection. Contrast with solid lesions, which need none.

## 14. Endocarditis / prosthetic joint (red X)

### WHAT YOU SEE

Heart and knee-joint with red Xs, 'ENDOCARDITIS / PROSTHETIC KNEE JOINT — none of those need prophylaxis', bottom right.

### WHAT IT ENCODES

GI endoscopy is NOT an indication for endocarditis prophylaxis (current guidelines dropped GI/GU procedures) and prosthetic joints do NOT require antibiotics for endoscopy. Prophylaxis is for biliary drainage/PEG/cystic-FNA, never these.

### BOARD TRAP

'Give antibiotics before colonoscopy/ERCP to prevent endocarditis or protect a prosthetic joint' → wrong; neither needs GI-endoscopy antibiotic prophylaxis.